Syllabus for Midterm exam – Function, Limit, Continuity, Differentiation, Application of differentiation, Taylor and Maclaurin series, Analysis of function, Indeterminate forms and $L'Hôpital$ rule.

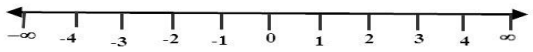
Textbook: Calculus– James Stewart - 8th edition
Reference Book: Calculus – Howard Anton- 10th edition

Introduction

Real Numbers, Intervals, and Inequalities

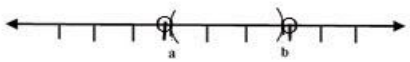
In mathematics, a real number is a value that represents a quantity along a line.

The real numbers include all the rational numbers, such as the **integer** −5 and the **fraction** 4/3, and all the **irrational numbers**, such as $\sqrt{2}$(1.41421356…, the **square root of 2**, an irrational algebraic number). Included within the irrationals are the transcendental numbers, such as π (3.14159265…). Real numbers can be thought of as points on an infinitely long line called the number line or real line, where the points corresponding to integers are equally spaced.

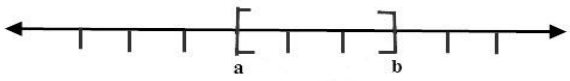


In calculus we will be concerned with sets of real numbers, called intervals, which correspond to line segments on a coordinate line. For example, if $a < b$, then the ***open interval*** from a to b, denoted by (a, b) is the line segment extending from a to b, excluding the end points; and the ***closed interval*** from a to b, denoted by $[a, b]$ is the line segment extending from a to b, including the end points. These sets can be expressed as

$(a, b) = \{x: a < x < b\}$ The open interval from a to b.

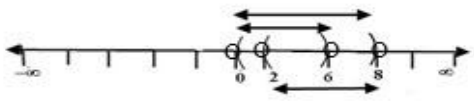


$[a, b] = \{x: a \leq x \leq b\}$ The closed interval from a to b.

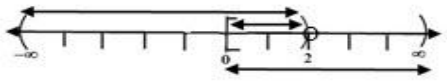


Unions and intersections of intervals: If A and B are two sets, then the **union** of A and B (denoted by $A \cup B$) is the set whose members belong to A or B (or both), and the **intersection** of A and B (denoted by $A \cap B$) is the set whose members belong to both$A$ and B. For example,

$$\{x: 0 < x < 6\} \cup \{x: 2 < x < 8\} = \{x: 0 < x < 8\}$$



$$\{x: x < 2\} \cap \{x: x \geq 0\} = \{x: 0 \leq x < 2\}$$



$$\{x: x < 2\} \cap \{x: x > 2\} = \emptyset$$



$$\{x: 0 < x < 6\} \backslash \{x: 2 \leq x < 8\} = \{x: 0 < x < 2\}$$



Or in interval notation,

$$(0, 6) \cup (2, 8) = (0, 8)$$

$$(-\infty, 2) \cap [0, +\infty) = [0, 2)$$

$$(-\infty, 2) \cap (2, +\infty) = \emptyset$$

$$(0, 6) \backslash [2, 8) = (0, 2)$$

Solving Inequalities

Theorem (Properties of Inequalities). Let *a, b, c*, and *d* be real numbers.

(i) If $a < b$ and $b < c$, then $a < c$.
(ii) If $a < b$, then $a + c < b + c$ and $a - c < b - c$.
(iii) If $a < b$, then $ac < bc$ when c is positive and $ac > bc$ when c is negative.
(iv) If $a < b$ and $c < d$, then $a + c < b + d$.
(v) If a and b are both positive or both negative and $a < b$, then $\frac{1}{a} > \frac{1}{b}$.

A **solution** of an inequality in an unknown *x* is a value of *x* that makes the inequality a true statement. For example, $x = 2$ is a solution of the inequality $x < 5$, but $x = 6$ is not. The set of all solutions of an inequality is called its solution set.

Example-1: Solve the inequality $3 + 14x \leq 2x - 9$.

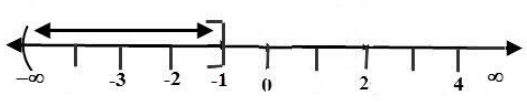
Solution: Given that,

$3 + 14x \leq 2x - 9$

$14x \leq 2x - 12$ (subtracted 3 from both sides)

$12x \leq -12$ (subtracted $2x$ from both sides)

$x \leq -1$ (multiplied both sides by $\frac{1}{12}$)



Example-2: Solve the inequality $7 \leq 2 - 5x < 12$.

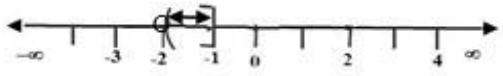
Solution: The given inequality is a combination of two inequalities.

$7 \leq 2 - 5x < 12$ (given)

$5 \leq -5x < 10$ (subtracted 2 from each member)

$-1 \geq x > -2$ (multiplied by $-\frac{1}{5}$ and reversed the inequality signs)

$-2 < x \leq -1$ (rewritten the inequalities with the smaller number on the left)



Example-3: Solve the inequality $x^2 - 3x > 10$.

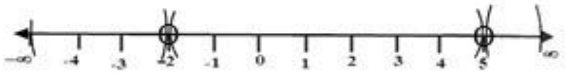
Solution: The given inequality is a combination of two inequalities.

$x^2 - 3x > 10$ (given)

$x^2 - 3x - 10 > 0$ (subtracted 10 from both sides)

$(x + 2)(x - 5) > 0$ (factorizing left side)

The values of x for which $x + 2 = 0$ or $x - 5 = 0$ are $x = -2$ and $x = 5$. These points divide the coordinate line into three open intervals, $(-\infty, -2), (-2, 5), (5, +\infty)$on each of which the product $(x + 2)(x - 5)$ has constant sign. To determine those signs we will choose an arbitrary point in each interval at which we will determine the sign; these are called test points. The result can be organized as follows:



Chapter 1

Function, Limits and Continuity

Function: A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E. Here the set D is the domain.

Some examples of finding domain:

1. Find the domain of each function.

(a) $f(x) = \sqrt{x+2}$.

Solution Because the square root of a negative number is not defined (as a real number), the domain of f consists of all values of x such that $x + 2 \geq 0$. This is equivalent to $x > -2$, so the domain is the interval $[-2, \infty)$.

(b) $g(x) = \frac{1}{x^2-x}$.

Solution: Snice, $g(x) = \frac{1}{x(x-1)}$ and division by 0 is not allowed, we see that $g(x)$ is not defined when $x = 0$ or $x = 1$. Thus, the domain of g is, $(-\infty, 0) \cup (0,1) \cup (1, \infty)$.

- More examples- 2,8 [**Ref.** James Stewart 8th edition, page-12,16]

Problem Set 1.1

Find the domain of each function.

1. $f(x) = \frac{x+4}{x^2-9}$

2. $f(x) = \frac{2x^3-5}{x^2+x-6}$

3. $f(t) = \sqrt[3]{2t-1}$

4. $g(t) = \sqrt{3-t} - \sqrt{2+t}$.

5. $f(x) = \frac{1}{\sqrt{x+7}}$

6. $f(x) = \frac{1}{\sqrt{5-2x}}$

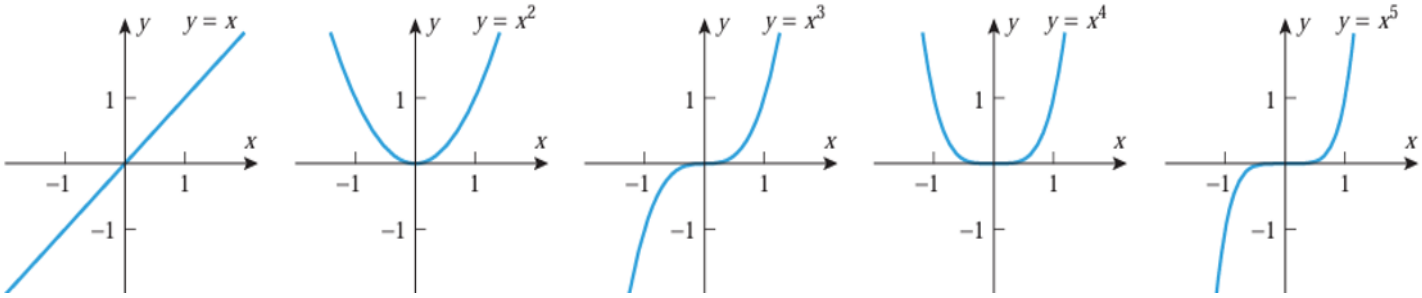
7. $f(x) = \frac{1}{x^2-7x+12}$

8. $h(t) = \sqrt{t+4} - \sqrt{2+t}$.

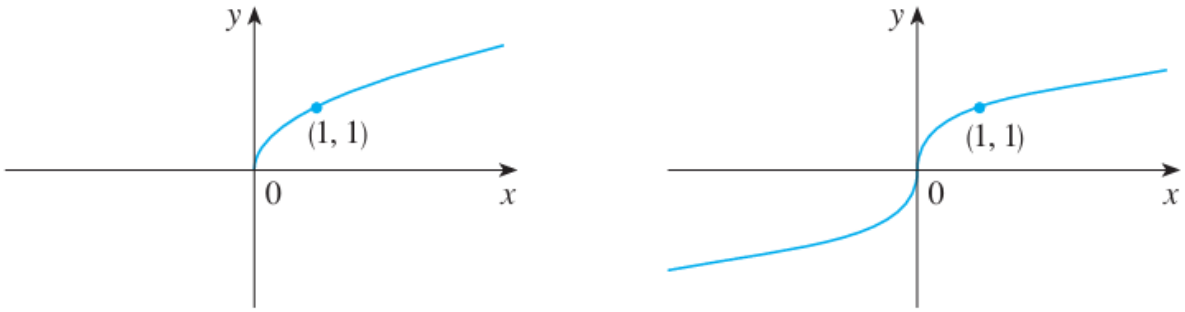
[**Ref.** James Stewart 8th edition, page-21]

Power function: A function of the form $f(x) = y = x^a$, where a is a constant, is called a power function.

(i) $a = n$, where n is a positive integer,

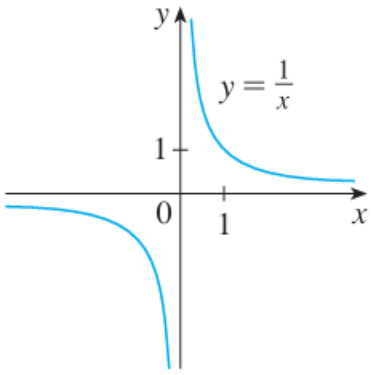


(ii) $a = \frac{1}{n}$, where n is a positive integer,



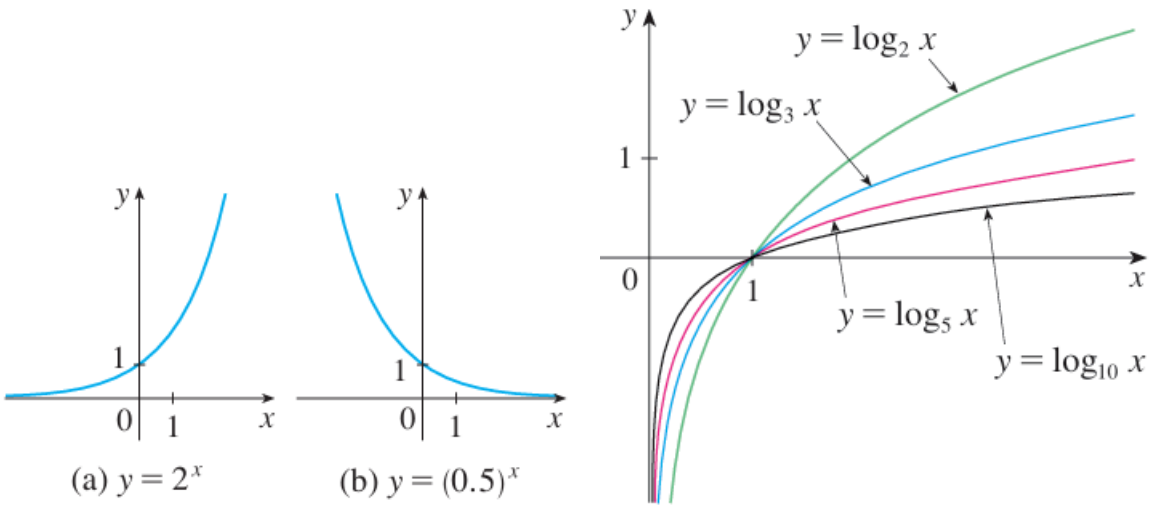
(a) $f(x) = \sqrt{x}$ (b) $f(x) = \sqrt[3]{x}$

(iii) **Reciprocal function:** $a = -1$,



Exponential & Logarithmic function:

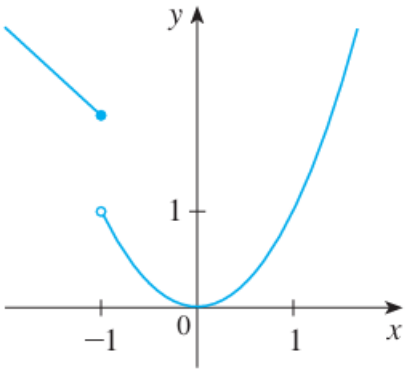
The exponential functions are the functions of the form $f(x) = b^x$. The logarithmic functions $f(x) = \log_b x$, where the base b is a positive constant, are the inverse functions of the exponential functions.



- More functions: Trigonometric function- $\sin x$, $\cos x$, Absolute value function- $y = |x|$.

  [**Ref.** James Stewart 8th edition, page-16,31]

Piecewise function: The functions are different formulas in different parts of their domains. Such functions are called piecewise defined functions. $f(x) = \begin{cases} 1 - x, & \text{if } x \leq -1 \\ x^2, & \text{if } x > -1 \end{cases}$.



New function from old function: Vertical and Horizontal Shifts:

- $y = f(x) + c$, shift the graph $y = f(x)$, vertically c units upward.
- $y = f(x) - c$, shift the graph $y = f(x)$, vertically c units downward.
- $y = f(x + c)$, shift the graph $y = f(x)$, horizontally c units to the left.
- $y = f(x - c)$, shift the graph $y = f(x)$, horizontally c units to the right.

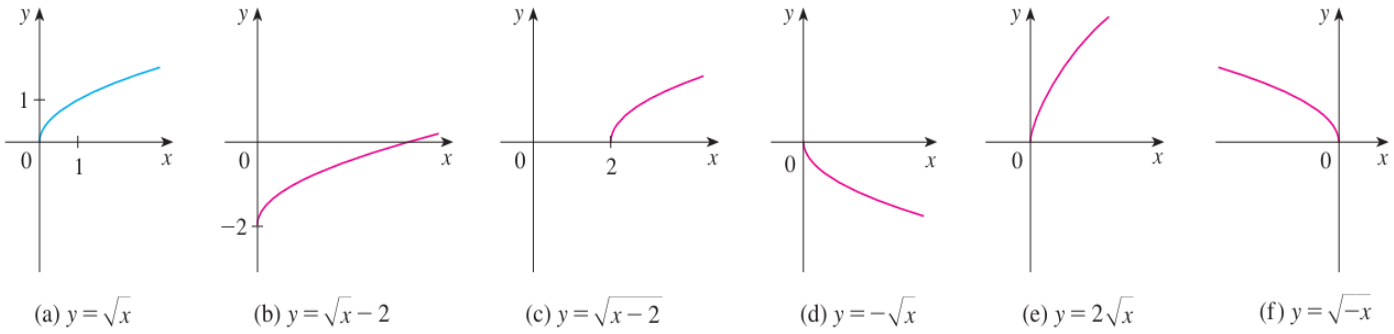
Vertical and Horizontal Stretching and Reflecting: $c > 1$.

- $y = cf(x)$, stretch the graph $y = f(x)$, vertically by a factor of c.
- $y = \frac{1}{c} f(x)$, shrink the graph $y = f(x)$, vertically by a factor of c.
- $y = f(cx)$, shrink the graph $y = f(x)$, horizontally by a factor of c.
- $y = f\left(\frac{x}{c}\right)$, stretch the graph $y = f(x)$, horizontally by a factor of c.
- $y = -f(x)$, reflect the graph of $y = f(x)$ about the $x - axis$.
- $y = f(-x)$, reflect the graph of $y = f(x)$ about the $y - axis$.

Example of function sketching:

1. Given the graph of $y = \sqrt{x}$, use transformations to graph $y = \sqrt{x} - 2$, $y = \sqrt{x - 2}$, $y = -\sqrt{x}$, $y = 2\sqrt{x}$ and $y = \sqrt{-x}$.

Solution: The graph of the square root function, $y = \sqrt{x}$, obtained from Fig- (ii) (a). In the other parts of the figure we sketch $y = \sqrt{x} - 2$ by shifting 2 units downward, $y = \sqrt{x - 2}$ by shifting 2 units to the right, $y = -\sqrt{x}$ by reflecting about the $x - axis$, $y = 2\sqrt{x}$ by stretching vertically by a factor of 2, and $y = \sqrt{-x}$ by reflecting about the $y - axis$.



(a) $y = \sqrt{x}$ (b) $y = \sqrt{x} - 2$ (c) $y = \sqrt{x - 2}$ (d) $y = -\sqrt{x}$ (e) $y = 2\sqrt{x}$ (f) $y = \sqrt{-x}$

- More examples: 2, 3, 11 [**Ref.** James Stewart 8th edition, page-38, 39, 62]

Problem Set 1.2

Sketch the graph.

1. $y = -x^2$

2. $y = (x-3)^2$

3. $y = x^3 + 1$

4. $f(x) = \ln(x+6)$

5. $f(x) = \ln(x+1)$

6. $y = x^2 - 2x + 2$

7. $f(x) = \begin{cases} -x, & if\ x < 0 \\ e^x - 1, & if\ x \geq 0 \end{cases}$

[**Ref.** James Stewart 8th edition, page 43,70]

8. $y = -(x+1)^2 - 3$

9. $y = 1 + \sqrt{x-4}$

10. $y = |x+2| - 2$

11. $y = 3 - \sqrt{x+1}$

12. $y = \sqrt{x^2 - 4x + 4}$

13. $y = \frac{1}{2}(x-3)^2 + 3$

[**Ref.** Howard Anton-10th edition, page-25]

14. $y = \sqrt{7-x}$

15. $y + 5 = |x|$

16. $f(x) = \ln(x-2)$

17. $x = y^2 - 1$

18. $y = x^2 - 2$

19. $y = \sqrt{x+5}$

20. $f(x) = \begin{cases} x^2, & if\ x < 2 \\ x+3, & if\ x \geq 2 \end{cases}$

Limits: Suppose $f(x)$ is defined when x is near the number a. (This means that f is defined on some open interval that contains a, except possibly at a itself.) Then we write $\lim_{x \to a} f(x) = L$ and say, "the limit of $f(x)$, as x approaches a, equals L".

- More properties (Power Law, Limit Law) [**Ref.** James Stewart 8th edition, page-95, 96, 191]

Examples of evaluating limits:

1. $\lim_{t \to 0} \frac{\sqrt{t^2+9}-3}{t^2}$

$= \lim_{t \to 0} \frac{\sqrt{t^2+9}-3}{t^2} \cdot \frac{\sqrt{t^2+9}+3}{\sqrt{t^2+9}+3}$

$= \lim_{t \to 0} \frac{(t^2+9)-9}{t^2(\sqrt{t^2+9}+3)}$

$= \lim_{t \to 0} \frac{t^2}{t^2(\sqrt{t^2+9}+3)}$

$= \lim_{t \to 0} \frac{1}{\sqrt{t^2+9}+3}$

$= \frac{1}{\sqrt{9}+3} = \frac{1}{6}$

2. $\lim_{x \to 1} \frac{x^2-6x+5}{x-1}$

$= \lim_{x \to 1} \frac{(x-5)(x-1)}{x-1}$

$= \lim_{x \to 1} (x-5) = -4$

3. $\lim_{x \to \infty} \frac{3x-2}{2x+1} = \lim_{x \to \infty} \frac{3-\frac{2}{x}}{2+\frac{1}{x}} = \frac{3}{2}$

- More examples: 5,6 [**Ref.** James Stewart 8th edition, page-99]

To remember: $\lim_{x \to 0} \frac{\sin x}{x} = 1$

Problem Set 1.3

Evaluate the limits.

1. $\lim_{x\to5}\frac{x^2-6x+5}{x-5}$
2. $\lim_{t\to-3}\frac{t^2-9}{2t^2+7t+3}$
3. $\lim_{h\to0}\frac{(-5+h)^2-25}{h}$
4. $\lim_{h\to0}\frac{(2+h)^3-8}{h}$
5. $\lim_{h\to0}\frac{\sqrt{9+h}-3}{h}$
6. $\lim_{u\to2}\frac{\sqrt{4u+1}-3}{u-2}$
7. $\lim_{t\to0}\frac{\sqrt{t+1}-\sqrt{1-t}}{t}$
8. $\lim_{x\to\infty}\frac{3x-2}{2x+1}$
9. $\lim_{x\to\infty}\frac{x^2}{\sqrt{x^4+1}}$
10. $\lim_{x\to-\infty}\frac{\sqrt{1+4x^6}}{2-x^3}$
11. $\lim_{x\to\infty}\frac{\sqrt{x+3x^2}}{4x-1}$
12. $\lim_{x\to\infty}\frac{x+3x^2}{4x-1}$
13. $\lim_{x\to\infty}\sqrt{9x^2+x}-3x$
14. $\lim_{x\to-\infty}\sqrt{4x^2+3x}+x$
15. $\lim_{x\to\infty}\frac{9x^2}{2x^4+1}$
16. $\lim_{x\to0}\frac{1-\cos4x}{2x}$
17. $\lim_{x\to\infty}\frac{x^3-2}{2x^4+1}$
18. $\lim_{x\to0}\frac{\sin3x}{\tan2x}$
19. $\lim_{x\to1}\frac{\sqrt{x}-1}{x-1}$
20. $\lim_{x\to0}\frac{\cos x}{\sin x-3}$

[**Ref.** James Stewart 8th edition, page- 102, 103, 138]

Two-Sided limits: $\lim_{x\to a}f(x)=L$ if and only if $\lim_{x\to a^+}f(x)=L=\lim_{x\to a^-}f(x)$.

Example-1: If $f(x)=\begin{cases}\sqrt{x-4}, & if\ x>4\\ 8-2x, & if\ x<4\end{cases}$, Determine whether $\lim_{x\to4}f(x)$ exists.

Solution: Since $f(x)=\sqrt{x-4}$ for $x>4$, we have, $\lim_{x\to4^+}f(x)=\lim_{x\to4^+}\sqrt{x-4}=\sqrt{4-4}=0$.

Since $f(x)=8-2x$ for $x<4$, we have, $\lim_{x\to4^-}f(x)=\lim_{x\to4^-}8-2x=8-2.4=0$.

The right- and left-hand limits are equal. Thus, the limit exists and $\lim_{x\to4}f(x)$.

- More examples: 7,8 [**Ref.** James Stewart 8th edition, page-100]

Problem Set 1.3.1

1. Let, $f(x)=\begin{cases}x^2+1, & x<1\\ (x-2)^2, & x\geq1\end{cases}$, a) Find $\lim_{x\to1^-}f(x)$ and $\lim_{x\to1^+}f(x)$. b) Does $\lim_{x\to1}f(x)$ exist?
2. Let, $B(t)=\begin{cases}4-\frac{1}{2}t, & t<2\\ \sqrt{t+c}, & t\geq2\end{cases}$, Find the value c so that $\lim_{t\to2}B(t)$ exists. [**Ref.** James Stewart 8th edition, page-103]
3. Let, $f(x)=\begin{cases}9-x^2, & x\geq-3\\ \frac{k}{x^2}, & x<-3\end{cases}$, Find the value k so that $\lim_{x\to-3}f(x)$ exists.
4. Let, $f(x)=\begin{cases}7x-2, & x\leq1\\ kx^2, & x>1\end{cases}$, Find the value k so that $\lim_{x\to1}f(x)$ exists. [**Ref.** Howard Anton 10th edition, page-119]

Continuity: A function f is continuous at a number a if

a) $f(a)$ is defined.

b) $\lim_{x\to a} f(x)$ exists, i.e. $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$.

c) $\lim_{x\to a} f(x) = f(a)$.

➢ More definition and examples: 2 (a), (b), (c) [**Ref.** James Stewart 8th edition, page-114]

Examples of checking continuity:

1: Test continuity of $f(x) = \begin{cases} x^2+1, & x<1 \\ x+1, & x\geq 1 \end{cases}$ at $x = 1$.

Solution: a) $f(x)$ is defined at $x = 1$

b) LHL: $\lim_{x\to 1^-} f(x) = (x^2+1)_{x=1} = 2$

RHL: $\lim_{x\to 1^+} f(x) = (x+1)_{x=1} = 2$

$\therefore \lim_{x\to 1} f(x)$ exists and $= 2$

c) $\lim_{x\to 1} f(x) = f(1) = 2$

$\therefore$ $f(x) =$ is continuous at $x = 1$.

2: Test continuity of $f(x) = \begin{cases} 2x+1, & x<1 \\ 4-2x, & x\leq 1 \end{cases}$ at $x = 1$.

Solution: a) $f(x)$ is defined at $x = 1$

b) LHL: $\lim_{x\to 1^-} f(x) = (2x+1)_{x=1} = 3$

RHL: $\lim_{x\to 1^+} f(x) = (4-2x)_{x=1} = 2$

c) $\therefore \lim_{x\to 1} f(x)$ does not exist

$\therefore$ $f(x)$ is not continuous at $x = 1$

Problem Set 1.4

Discuss the continuity of the function and sketch it.

[1-7] [**Ref.** James Stewart 8th edition, page-125]

1. $f(x) = \begin{cases} 1-x^2 & \text{if } x\leq 1 \\ \ln x & \text{if } x>1 \end{cases}$ at $x = 1$

2. $f(x) = \begin{cases} \cos x & \text{if } x < \frac{\pi}{4} \\ \sin x & \text{if } x \geq \frac{\pi}{4} \end{cases}$ at $x = \frac{\pi}{4}$

3. $f(x) = \begin{cases} x^2, & x<-1 \\ x, & -1\leq x\leq 1 \\ \frac{1}{x}, & x>1 \end{cases}$ at $x = 1$

4. $f(x) = \begin{cases} 2^x & \text{if } x\leq 1 \\ 3-x & \text{if } 1\leq x\leq 4 \\ \sqrt{x} & \text{if } x>4 \end{cases}$ at $x = 4$

5. $f(x) = \begin{cases} x+2 & \text{if } x<0 \\ e^x & \text{if } 0\leq x\leq 1 \\ 2-x & \text{if } x>1 \end{cases}$ at $x = 0$

6. $f(x) = \begin{cases} 4-x^2, & x<0 \\ 4+x, & x\geq 0 \end{cases}$ at $x = 0$

7. $f(x) = \begin{cases} x^2-2x+1, & x\leq 2 \\ 3x-2, & x>2 \end{cases}$ at $x = 2$

8. $f(x) = \begin{cases} x^2-2\cos x-1, & x\leq 0 \\ \sec x-4, & x>0 \end{cases}$ at $x = 0$

9. $f(x) = \begin{cases} 2x+1 & \text{if } x<1 \\ 4-2x & \text{if } x\geq 1 \end{cases}$ at $x = 1$

10. $f(x) = \begin{cases} 5-2x & \text{if } x<-3 \\ x^2+2 & \text{if } x\geq -3 \end{cases}$ at $x = -3$

➢ More exercise: 17-19 [**Ref.** James Stewart 8th edition, page-12]

Chapter 2

Derivatives

First Principle of Differentiation: The derivative of a function *f* at *x*, denoted by $f'(x)$, is

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ if this limit exists.

Table of Differentiation

1. $\frac{d}{dx}(c) = 0$, where c is a constant.	2. $\frac{d}{dx}(x) = 1$
3. $\frac{d}{dx}(x^n) = nx^{n-1}$	4. $\frac{d}{dx}(e^x) = e^x$
5. $\frac{d}{dx}(a^x) = a^x \ln a$	6. $\frac{d}{dx}(\log_a x) = \frac{1}{x}\log_a e$
7. $\frac{d}{dx}(\sin x) = \cos x$	8. $\frac{d}{dx}(\cos x) = -\sin x$
9. $\frac{d}{dx}(\tan x) = \sec^2 x$	10. $\frac{d}{dx}(\sec x) = \sec x \tan x$
11. $\frac{d}{dx}(\cot x) = -\csc^2 x$	12. $\frac{d}{dx}(\csc x) = -\csc x \cot x$
13. $\frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$	14. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
15. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$	16. $\frac{d}{dx}(\sinh x) = \cosh x$
17. $\frac{d}{dx}(\cosh x) = \sinh x$	18. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Derivative of a Constant times a Function: If f is differentiable at *x* and *c* is any real number, then *cf* is also differentiable at *x* and

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$$

Derivatives of Sums and Differences: If *f* and *g* are differentiable at *x,* then so are $f + g$ and $f - g$,

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x)] - \frac{d}{dx}[g(x)]$$

The Product Rule: If *f* and *g* are differentiable at *x,* then

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$$

The Quotient Rule: If *f* and *g* are differentiable at *x* and $g(x) \neq 0$, then

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f\ o\ g$ is differentiable at x. Moreover, if

$$y = f\big(g(x)\big) \text{ and } u = g(x)$$

then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Implicit Differentiation

The derivative of the function of the form $f(x, y) = 0$ can be obtained by differentiating each term separately and solving for $\frac{dy}{dx}$. This technique of differentiation is known as Implicit Differentiation.

Suppose, $\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2)\frac{dy}{dx} = 2y\frac{dy}{dx} = 2yy'$

Parametric Differentiation

Suppose that a curve is given by the parametric equation $x = f(t)$ and $y = g(t)$ where $f(t)$ and $g(t)$ are both differentiable. Then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Some workout examples:

Example-1: Find the derivative of the function $f(x) = x^2 - 8x + 9$ using first principle of differentiation.

Solution: First principle of differentiation states that

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{[(x+h)^2 - 8(x+h) + 9] - [x^2 - 8x + 9]}{h} \\
&= \lim_{h\to 0} \frac{x^2 + 2hx + h^2 - 8x - 8h + 9 - x^2 + 8x - 9}{h} \\
&= \lim_{h\to 0} \frac{2hx + h^2 - 8h}{h} \\
&= \lim_{h\to 0} \frac{h(2x + h - 8)}{h} \\
&= \lim_{h\to 0} (2x + h - 8) \\
&= 2x - 8
\end{aligned}$$

Example-2: Find the derivative of the function $f(x) = \sin x$ using first principle of differentiation.

Solution:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h)-\sin x}{h}$$

$$= \lim_{h\to 0} \frac{2\cos\frac{x+h+x}{2}\sin\frac{x+h-x}{2}}{h}$$

$$= \lim_{h\to 0} \frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h}$$

$$= 2\times\frac{1}{2}\times\lim_{h\to 0}\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\times\lim_{h\to 0}\cos\left(x+\frac{h}{2}\right) = \cos x \quad [\text{Since}\lim_{x\to 0}\frac{\sin x}{x} = 1]$$

Example-3: Differentiate (a) $y = \sin(x^2)$ and (b) $y = \sin^2 x$.

Solution:

(a) $\frac{dy}{dx} = \frac{d}{dx}\sin(x^2)$ $= \cos(x^2)\times\frac{d}{dx}x^2$ $= 2x\cos(x^2)$	(b) $\frac{dy}{dx} = \frac{d}{dx}\sin^2 x$ $= \frac{d}{dx}(\sin x)^2$ $= 2\sin x\times\frac{d}{dx}\sin x = 2\sin x\cos x = \sin 2x$

Example-4: Differentiate (a) $y = (1-2x)^6$ and (b) $y = \sqrt{x^2+1}$

Solution:

(a) $\frac{dy}{dx} = \frac{d}{dx}(1-2x)^6$ $= 6(1-2x)^5\times\frac{d}{dx}(1-2x)$ $= 6(1-2x)^5(-2)$ $= -12(1-2x)^5$	(b) $\frac{dy}{dx} = \frac{d}{dx}\sqrt{x^2+1} = \frac{d}{dx}(x^2+1)^{1/2}$ $= \frac{1}{2}(x^2+1)^{-\frac{1}{2}}\frac{d}{dx}(x^2+1)$ $= \frac{1}{2}(x^2+1)^{-\frac{1}{2}}\times 2x = \frac{x}{\sqrt{x^2+1}}$

Example-5: Differentiate (a) $y = x^3e^{2x}$ and (b) $y = \frac{\ln x}{x^2}$.

Solution:

(a) $\frac{dy}{dx} = \frac{d}{dx}x^3e^{2x}$ $= x^3\frac{d}{dx}e^{2x} + e^{2x}\frac{d}{dx}x^3$ $= 2x^3e^{2x} + 3x^2e^{2x}$	(b) $\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\ln x}{x^2}\right)$ $= \frac{x^2\frac{d}{dx}\ln x - \ln x\frac{d}{dx}x^2}{(x^2)^2}$ $= \frac{x^2\times\frac{1}{x} - \ln x\times 2x}{x^4} = \frac{x-2x\ln x}{x^4}$

Example-6: Differentiate (a) $y = x^{sin\,x}$ and (b) $y = \ln(\tan^{-1}(3x))$.

Solution:

(a) $y = x^{sin\,x}$ Taking logarithm on both sides, $\ln y = \ln\left(x^{\sin x}\right) = \sin x \ln x$ Differentiating with respect to x, $\frac{1}{y}\frac{dy}{dx} = \cos x \ln x + \sin x . \frac{1}{x}$ *or*, $\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$	(b) $\frac{dy}{dx} = \frac{d}{dx}\ln(\tan^{-1}(3x))$ $= \frac{1}{\tan^{-1}(3x)} \times \frac{3}{1+9x^2}$

Example-7: Given, $x^2y + 3xy^3 - x = 3$. Find $\frac{dy}{dx}$ by implicit differentiation.

Solution: Differentiating with respect to x,

$$x^2\frac{dy}{dx} + 2xy + 9xy^2\frac{dy}{dx} + 3y^3 - 1 = 0$$

or, $(x^2 + 9xy)\frac{dy}{dx} = 1 - 2xy$

or, $\frac{dy}{dx} = \frac{1-2xy}{x^2+9xy}$

Example-8: Given, $x = t^3 - 3t,\ y = t^3 - 3t^2$. Find $\frac{dy}{dx}$ by parametric differentiation.

Solution: Differentiating x with respect to t, $\frac{dx}{dt} = 3t^2 - 3$.

Differentiating y with respect to t, $\frac{dy}{dt} = 3t^2 - 6t$.

Therefore, $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2-6t}{3t^2-3}$

Problem Set 2.1
Product Rule and Quotient Rule

1-27 Differentiate. (Ref: Calculus- James Stewart- 8th Edition. Page# 188)

1. $f(x) = (3x^2 - 5x)e^x$
2. $y = \frac{x}{e^x}$
3. $g(x) = \left(x + 2\sqrt{x}\right)e^x$
4. $y = \frac{e^x}{1-e^x}$
5. $g(x) = \frac{1+2x}{3-4x}$
6. $G(x) = \frac{x^2-2}{2x+1}$
7. $H(u) = \left(u - \sqrt{u}\right)\left(u + \sqrt{u}\right)$
8. $f(x) = e^x \cos x$
9. $f(x) = x \cos x + 2 \tan x$
10. $y = 2 \sec x - \operatorname{cosec} x$
11. $y = \sec x \tan x$
12. $g(\theta) = e^\theta(\tan\theta - \theta)$
13. $y = c \cos t + t^2 \sin t$
14. $f(t) = \frac{\cot t}{e^t}$
15. $J(v) = (v^3 - 2v)(v^{-4} + v^{-2})$
16. $F(y) = \left(\frac{1}{y^2} - \frac{3}{y^4}\right)(y + 5y^3)$
17. $f(z) = (1 - e^z)(z + e^z)$
18. $y = \frac{x^2+1}{x^3-1}$
19. $y = \frac{\sqrt{x}}{2+x}$
20. $y = \frac{t^3+3t}{t^2-4t+3}$
21. $f(x) = x^2 \sin x$
22. $y = \frac{x}{2-\tan x}$
23. $y = \sin\theta \cos\theta$
24. $f(\theta) = \frac{sin\theta}{1+\cos\theta}$
25. $y = \frac{\cos x}{1-\sin x}$
26. $y = \frac{t \sin t}{1+t}$
27. $y = e^x \cosh x$

28-35 Differentiate. (Ref: Calculus- Howard Anton- 10th Edition. Page# 147)

28. $y = (x^2 + 1)\sec x$
29. $f(x) = \frac{\sin x \sec x}{1+x\tan x}$
30. $s = (1 + t)\sqrt{t}$
31. $y = \frac{x^3+2x^2-1}{x+5}$
32. $f(x) = \left(2\sqrt{x} + 1\right)\frac{(2-x)}{(x^2+3x)}$
33. $y = (2x^7 - x^2)\left(\frac{x-1}{x+1}\right)$
34. $f(x) = \left(\frac{1}{x} + \frac{1}{x^2}\right)(3x^3 + 27)$
35. $f(x) = \left(\frac{3x+2}{x}\right)(x^{-5} + 1)$

Chain Rule

1-17 Differentiate. (Ref: Calculus- James Stewart- 8th Edition. Page# 198, 214, 264)

1. $f(x) = (5x^6 + 2x^3)^4$
2. $y = x^2e^{-3x}$
3. $f(x) = (1 + x + x^2)^{99}$
4. $y = \frac{1}{\sin^{-1}x}$
5. $f(x) = \sqrt{5x + 1}$
6. $y = x\tan^{-1}\sqrt{x}$
7. $f(x) = \frac{1}{\sqrt[3]{x^2-1}}$
8. $y = \tanh\sqrt{x}$

9. $f(\theta) = \cos(\theta^2)$
10. $g(\theta) = \cos^2\theta$
11. $f(t) = t\sin\pi t$
12. $f(t) = e^{at}\sin bt$
13. $g(x) = e^{x^2-x}$
14. $y = \sinh^2 x$
15. $y = \sinh(x^2)$
16. $y = \ln(\sinh t)$
17. $y = \sinh(\ln t)$

18-30 Differentiate. (Ref: Calculus- Howard Anton- 10th Edition. Page# 158)

18. $y = (3x^2 + 2x - 1)^6$
19. $f(x) = \frac{1}{(x^5-x+1)^9}$
20. $f(x) = 4x + 5\sin^4 x$
21. $f(x) = \tan^4(x^3)$
22. $f(x) = \cos^3\left(\frac{x}{x+1}\right)$
23. $y = \sqrt{\cos(5x)}$
24. $y = 2\sec^2(x^7)$
25. $y = x\tan\left(\frac{1}{x}\right)$
26. $f(x) = \sin\left(\frac{1}{x^2}\right)$
27. $f(x) = \sqrt[3]{12 + \sqrt{x}}$
28. $f(x) = \cos(\cos x)$
29. $f(x) = \sin(\tan 3x)$
30. $f(x) = (x^2 + x)^5\sin^8 x$

Logarithmic Differentiation

1-9 Find $\frac{dy}{dx}$. (Ref: Calculus- James Stewart- 8th Edition. Page# 223)

1. $y = \sqrt{\frac{x-1}{x^4+1}}$
2. $y = \sqrt{x^x}$
3. $y = \sqrt{x}e^{x^2-x}(x+1)^{\frac{2}{3}}$
4. $y = (\cos x)^x$
5. $y = x^x$
6. $y = (\sin x)^{\ln x}$
7. $y = x^{\cos x}$
8. $y = (\tan x)^{\frac{1}{x}}$
9. $y = (\ln x)^{\cos x}$

10-12 Find $\frac{dy}{dx}$. (Ref: Calculus- Howard Anton- 10th Edition. Page# 423, 425)

10. $y = \frac{x^2\sqrt[3]{7x-14}}{(1+x^2)^4}$
11. $y = x\sqrt[3]{1 + x^2}$
12. $y = \sqrt[5]{\frac{x-1}{x+1}}$

Implicit Differentiation

1-10 Find $\frac{dy}{dx}$ using implicit differentiation. (Ref: Calculus- James Stewart- 8th Edition. Page# 215)

1. $x^2 - 4xy + y^2 = 4$
2. $xe^y = x - y$
3. $x^4 + x^2y^2 + y^3 = 5$
4. $2x^2 + xy - y^2 = 2$
5. $x^3 - xy^2 + y^3 = 1$
6. $\frac{x^2}{x+y} = y^2 + 1$
7. $y\cos x = x^2 + y^2$
8. $\cos(xy) = 1 + \sin y$
9. $\sqrt{x+y} = x^4 + y^4$
10. $e^y \sin x = x + xy$

11-17 Find $\frac{dy}{dx}$ using implicit differentiation. (Ref: Calculus- Howard Anton- 10th Edition. Page# 163, 164, 166)

11. $5y^2 + \sin y = x^2$
12. $x^3 + y^3 = 3xy$
13. $\sin(x^2y^2) = x$
14. $x^2 = \frac{x+y}{x-y}$
15. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$
16. $\cos(xy^2) = y$
17. $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = 1$

Parametric Differentiation

1-6 Find $\frac{dy}{dx}$ using parametric differentiation. (Ref: Calculus- James Stewart- 8th Edition. Page# 655)

1. $x = t^2 + 1,\ y = t^2 + t$
2. $x = t^3 + 1,\ y = t^2 - t$
3. $x = e^t,\ y = te^{-t}$
4. $x = t^2 + 1,\ y = e^t - 1$
5. $x = t - \ln t,\ y = t + \ln t$
6. $x = \cos t,\ y = \sin 2t,\ 0 < t < \pi$

7-11 (Ref: Calculus- Howard Anton- 10th Edition. Page# 703, 704)

7. $x = \sqrt{t},\ y = 2t + 7$
8. $x = \sec t,\ y = \tan t$
9. $x = t + \cos t,\ y = 1 + \sin t$
10. $x = \sinh t,\ y = \cosh t$
11. $x = \frac{1}{2}t^2 + 1,\ y = \frac{1}{3}t^3 - t$

L'Hospital's Rule

L'Hospital's Rule tells us that if we have an indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ all we need to do is differentiate the numerator and differentiate the denominator and then take the limit.

Indeterminate Forms of Type $0/0$

A limit of the form $\lim_{x\to a} \frac{f(x)}{g(x)}$ in which $f(x) \to 0$ and $g(x) \to 0$ as $x \to a$ is called an **indeterminate form of type** $\mathbf{\frac{0}{0}}$. If $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Example-1: Find $\lim_{x\to 1} \frac{\ln x}{x-1}$.

Solution: The numerator and denominator have a limit of 0, so the limit is an indeterminate form of type $0/0$. Applying L'Hôpital's rule yields

$$\lim_{x\to 1} \frac{\ln x}{x-1} = \lim_{x\to 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)} = \lim_{x\to 1} \frac{\frac{1}{x}}{1}$$

$$= \lim_{x\to 1} \frac{1}{x} = 1$$

Example-2: Find $\lim_{x\to 0} \frac{\tan x - x}{x^3}$.

Solution: Noting that both $(\tan x - x) \to 0$ and $x^3 \to 0$ as $x \to 0$, so the limit is an indeterminate form of type $0/0$. Applying L'Hôpital's rule yields

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2} \text{ [Again } 0/0 \text{ form]}$$

$$= \lim_{x\to 0} \frac{2\sec^2 x \tan x}{6x} \text{ [Applying L'Hôpital's rule]}$$

$$= \frac{1}{3} \lim_{x\to 0} \sec^2 x \times \lim_{x\to 0} \frac{\tan x}{x} = \frac{1}{3} \times 1 \times 1 = \frac{1}{3}.$$

Indeterminate Forms of Type ∞/∞

The limit of a ratio, $f(x)/g(x)$, in which the numerator has limit ∞ and the denominator has limit ∞ is called an **indeterminate form of type** ∞/∞.

Example-3: Find $\lim_{x\to+\infty} \frac{x}{e^x}$.

Solution: The numerator and denominator have a limit of ∞, so the limit is an indeterminate form of type ∞/∞. Applying L'Hôpital's rule yields

$$\lim_{x\to+\infty}\frac{x}{e^x}=\lim_{x\to+\infty}\frac{1}{e^x}$$
$$=\frac{1}{e^\infty}=0$$

Example-4: Find $\lim_{x\to0^+}\frac{\ln x}{\csc x}$.

Solution: The numerator and denominator have a limit of ∞, so the limit is an indeterminate form of type ∞/∞. Applying L'Hôpital's rule yields

$$\lim_{x\to0^+}\frac{\ln x}{\csc x}=\lim_{x\to0^+}\frac{1/x}{-\csc x\cot x}$$
$$=\lim_{x\to0^+}\left(-\frac{\sin x}{x}\times\tan x\right)$$
$$=-\lim_{x\to0^+}\frac{\sin x}{x}\cdot\lim_{x\to0^+}\tan x$$
$$=-1\cdot0=0$$

Indeterminate Forms of Type $0\cdot\infty$

Thus far we have discussed indeterminate forms of type $0/0$ and ∞/∞. However, these are not the only possibilities. For example, the limit $\lim_{x\to0^+}(x\ln x)$ is an **indeterminate form of type $0\cdot\infty$** because the limit of the first factor is 0, the limit of the second factor is $-\infty$. Indeterminate form of type $0\cdot\infty$ can sometimes be evaluated by rewriting the product as a ratio, and then applying L'Hôpital's rule for indeterminate forms of type $0/0$ or ∞/∞.

Example-5: Find $\lim_{x\to0^+}x\ln x$.

Solution: This limit is an indeterminate form of type $0\cdot\infty$. We can rewrite the limit as $\lim_{x\to0^+}\frac{x}{1/\ln x}$ or $\lim_{x\to0^+}\frac{\ln x}{1/x}$.

However, the second form is the preferred initial choice because the derivative of $1/x$ is less complicated than the derivative of $1/\ln x$. Therefore,

$$\lim_{x\to0^+}x\ln x=\lim_{x\to0^+}\frac{\ln x}{1/x}$$
$$=\lim_{x\to0^+}\frac{1/x}{-1/x^2}$$
$$=\lim_{x\to0^+}(-x)=0$$

Indeterminate Forms of Type $\infty-\infty$

A limit problem that leads to one of the following expressions

$$(+\infty)-(+\infty),\qquad(-\infty)-(-\infty),$$
$$(+\infty)+(-\infty),\qquad(-\infty)+(+\infty)$$

is called an **indeterminate form of type $\infty-\infty$.** Indeterminate forms of type $\infty-\infty$ can sometimes be evaluated by combining the terms and manipulating the result to produce an indeterminate form of type $0/0$ or ∞/∞.

Example-6: Find $\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{\sin x}\right)$.

Solution: This limit is an indeterminate form of type $\infty-\infty$. Combining the two terms yields,

$$\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{\sin x}\right)=\lim_{x\to 0^+}\frac{\sin x-x}{x\sin x}\ [\text{Indeterminate form of type } 0/0]$$

$$=\lim_{x\to 0^+}\frac{\cos x-1}{\sin x+x\cos x}\ [\text{Again } 0/0 \text{ form}]$$

$$=\lim_{x\to 0^+}\frac{-\sin x}{\cos x+\cos x-x\sin x}=\frac{0}{2}=0$$

Indeterminate Forms of Type $0^0, \infty^0, 1^\infty$

Indeterminate forms of type $0^0, \infty^0, 1^\infty$ can sometimes be evaluated by first introducing a dependent variable $y=f(x)^{g(x)}$ and then computing the limit of $\ln y$. Since

$$\ln y=\ln\left[f(x)^{g(x)}\right]=g(x)\cdot\ln[f(x)]$$

The limit of $\ln y$ will be an indeterminate form of type $0\cdot\infty$, which can be evaluated by methods we have already studied.

Example-7: Find $\lim_{x\to 0}(1+\sin x)^{1/x}$.

Solution: This limit is an indeterminate form of type 1^∞. We begin by introducing a dependent variable

$$y=(1+\sin x)^{1/x}$$

and taking natural logarithm on both sides:

$$\ln y=\ln(1+\sin x)^{1/x}=\frac{1}{x}\ln(1+\sin x)=\frac{\ln(1+\sin x)}{x}$$

Thus,

$$\lim_{x\to 0}\ln y=\lim_{x\to 0}\frac{\ln(1+\sin x)}{x}\ [0/0 \text{ form}]$$

$$=\lim_{x\to 0}\frac{\cos x/(1+\sin x)}{1}=1$$

Since we have shown that $\ln y\to 1$ as $x\to 0$, implies that $y\to e$ as $x\to 0$. Thus,

$$\lim_{x\to 0}(1+\sin x)^{1/x}=e$$

Problem Set 2.2

1-35 Find the limit using L'Hospital's Rule. If L'Hospital's Rule doesn't apply, explain why. (Ref: Calculus- James Stewart- 8th Edition. Page# 311, 312 and Calculus- Howard Anton- 10th Edition. Page# 448)

1. $\lim_{x\to\left(\frac{\pi}{2}\right)^+} \frac{\cos x}{1-\sin x}$
2. $\lim_{x\to 0} \frac{\tan 3x}{\sin 2x}$
3. $\lim_{x\to\infty} \frac{e^x}{x^2}$
4. $\lim_{x\to 0} \frac{x^2}{1-\cos x}$
5. $\lim_{t\to 0} \frac{e^{2t}-1}{\sin t}$
6. $\lim_{\theta\to\pi} \frac{1+\cos\theta}{1-cos\theta}$
7. $\lim_{\theta\to\frac{\pi}{2}} \frac{1-\sin\theta}{1+\cos 2\theta}$
8. $\lim_{x\to\infty} \frac{x+x^2}{1-2x^2}$
9. $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$
10. $\lim_{x\to\infty} \frac{\ln\sqrt{x}}{x^2}$
11. $\lim_{x\to 0^+} \frac{\ln x}{x}$
12. $\lim_{u\to\infty} \frac{e^{\frac{u}{10}}}{u^3}$
13. $\lim_{t\to 1} \frac{t^8-1}{t^5-1}$
14. $\lim_{x\to 0} \frac{e^x-1-x}{x^2}$
15. $\lim_{t\to 0} \frac{8^t-5^t}{t}$
16. $\lim_{x\to 1} \frac{x^3-2x^2+1}{x^3-1}$
17. $\lim_{x\to\frac{1}{2}} \frac{6x^2+5x-4}{4x^2+16x-9}$
18. $\lim_{x\to 0} \frac{x-\sin x}{x-\tan x}$
19. $\lim_{x\to 0} \frac{\sinh x-x}{x^3}$
20. $\lim_{x\to\infty} x\sin\left(\frac{\pi}{x}\right)$
21. $\lim_{x\to 0} \frac{\tanh x}{\tan x}$
22. $\lim_{x\to\infty} x^3 e^{-x^2}$
23. $\lim_{x\to\infty} \sqrt{x}e^{-\frac{x}{2}}$
24. $\lim_{x\to 0} (\csc x - \cot x)$
25. $\lim_{x\to 1} \left(\frac{x}{x-1} - \frac{1}{\ln x}\right)$
26. $\lim_{x\to 0^+} \left(\frac{1}{x} - \frac{1}{\tan^{-1} x}\right)$
27. $\lim_{x\to 0^+} \left(\frac{1}{x} - \frac{1}{e^x-1}\right)$
28. $\lim_{x\to 0^+} x^{\sqrt{x}}$
29. $\lim_{x\to\infty} (x - \ln x)$
30. $\lim_{x\to 0} (1-2x)^{\frac{1}{x}}$
31. $\lim_{x\to 0^+} (\tan 2x)^x$
32. $\lim_{x\to 1^+} x^{\left(\frac{1}{1-x}\right)}$
33. $\lim_{x\to\infty} \left(1+\frac{a}{x}\right)^{bx}$
34. $\lim_{x\to\infty} x^{\frac{1}{x}}$
35. $\lim_{x\to 0^+} (4x+1)^{\cot x}$

Chapter 3

Application of Differentiation

Rates of Change

Example-1: Suppose that x and y are differentiable functions of t and are related by the equation $x^2 + y^2 = 2x + 4y$. (a) Given that $\frac{dx}{dt} = -5$, find $\frac{dy}{dt}$ when $(x, y) = (3,1)$. (b) Given that $\frac{dy}{dt} = 6$, find $\frac{dx}{dt}$ when $(x, y) = (1 + \sqrt{2}, 2 + \sqrt{3})$.

Solution: Given, $x^2 + y^2 = 2x + 4y$. Since x and y both are functions of t, differentiating the equation with respect to t, $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2\frac{dx}{dt} + 4\frac{dy}{dt}$ (i)

(a) Given, $\frac{dx}{dt} = -5$. We need to find $\frac{dy}{dt}$ when $x = 3$ and $y = 1$. Putting these values in (i), we have,

$$2 \cdot 3 \cdot (-5) + 2 \cdot 1 \cdot \frac{dy}{dt} = 2 \cdot (-5) + 4\frac{dy}{dt}$$

$$\text{or, } 2\frac{dy}{dt} - 4\frac{dy}{dt} = -10 + 30$$

$$\text{or, } \frac{dy}{dt} = -10$$

(b) Given, $\frac{dy}{dt} = 6$. We need to find $\frac{dx}{dt}$ when $x = 1 + \sqrt{2}$ and $y = 2 + \sqrt{3}$. Putting these values in (i),

$$2\left(1 + \sqrt{2}\right)\frac{dx}{dt} + 2\left(2 + \sqrt{3}\right) \cdot 6 = 2\frac{dx}{dt} + 4 \cdot 6$$

$$\text{or, } 2\frac{dx}{dt} + 2\sqrt{2}\frac{dx}{dt} - 2\frac{dx}{dt} = 24 - 24 - 12\sqrt{3}$$

$$\text{or, } \frac{dx}{dt} = -\frac{6\sqrt{3}}{\sqrt{2}}$$

Example-2: Let A be the area of a square whose sides have length x, and assume that x varies with the time t. At a certain instant the sides are 3 ft long and increasing at a rate of 2 ft/s. How fast is the area increasing at that instant?

Solution: We know the rate at which the sides of the square are increasing, and we want to find the rate at which the area is increasing at the instant when $x = 3$ ft; that is, we want to find

$$\left.\frac{dA}{dt}\right|_{x=3} \text{ given that } \frac{dx}{dt} = 2 \text{ ft/s}$$

A is the area of a square of sides x, so $A = x^2$.

Differentiating with respect to t yields

$$\frac{dA}{dt} = 2x\frac{dx}{dt}$$

Thus, when $x = 3$ ft, the area of the square is increasing at the rate of

$$\left.\frac{dA}{dt}\right|_{x=3} = 2 \times 3 \times 2 = 12 \text{ ft}^2\text{/s}.$$

Problem Set 3.1

1-9 (Ref: Calculus- Howard Anton- 10th Edition. Page# 168, 173, 175 and Calculus- James Stewart- 8th Edition. Page# 148, 149)

1. A ball is thrown into the air with a velocity of 40 ft/s, its height (in feet) after t seconds is given by $y = 40t - 16t^2$. Find the velocity and acceleration when $t = 2$. Ans: $v = -24$ ft/s, $a = -32$ ft/s^2
2. The displacement (in meters) of a particle moving in a straight line is given by the equation $s = \frac{1}{t^2}$, where t is measured in seconds. Find the velocity and acceleration of the particle at time $t = a, t = 1, t = 2,$ and $t = 3$. Ans: At $t = a, v = -\frac{2}{a^3}$ m/s, $a = \frac{6}{a^4}$ m/s^2; At $t = 1, v = -2$ m/s, $a = 6$ m/s^2; At $t = 2,, v = -\frac{1}{4}$ m/s, $a = \frac{3}{8}$ m/s^2; At $t = 3, v = -\frac{2}{27}$ m/s, $a = \frac{2}{27}$ m/s^2;
3. Suppose that x and y are differentiable functions of t and are related by the equation $y = x^3$. Find $\frac{dy}{dt}$ at time $t = 1$ if $x = 2$ and $\frac{dx}{dt} = 4$ at time $t = 1$. Ans: $\frac{dy}{dt} = 48$
4. Assume that oil spilled from a ruptured tanker spreads in a circular pattern whose radius increases at a constant rate of 2 ft/s. How fast is the area of the spill increasing when the radius of the spill is 60 ft? Ans: $\frac{dA}{dt} = 240\pi$ ft^2/s
5. Given that $4x^2 + 9y^2 = 1$.
   (a) If $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$ when $(x, y) = \left(\frac{1}{2\sqrt{2}}, \frac{1}{3\sqrt{2}}\right)$. Ans: $\frac{dy}{dt} = -2$
   (b) If $\frac{dy}{dt} = 8$, find $\frac{dx}{dt}$ when $(x, y) = \left(\frac{1}{3}, -\frac{\sqrt{5}}{9}\right)$. Ans: $\frac{dx}{dt} = 6\sqrt{5}$
6. Let V be the volume of a cylinder having height h and radius r, and assume that h and r vary with time. At a certain instant, the height is 6 in and increasing at 1 in/s, while the radius is 10 in and decreasing at 1 in/s. How fast is the volume changing at that instant? Is the volume increasing or decreasing at that instant? Ans: $\frac{dV}{dt} = 220\pi$ in^3/s, increasing
7. Oil spilled from a ruptured tanker spreads in a circle whose area increases at a constant rate of 6 ft^2/s. How fast is the radius of the spill increasing when the area is 9 ft^2? Ans: $\frac{dr}{dt} = \frac{1}{\sqrt{\pi}}$ ft/s
8. A particle is moving along the curve whose equation is $\frac{xy^3}{1+y^2} = \frac{8}{5}$. Assume that the x-coordinate is increasing at the rate of 6 units/s when the particle is at the point $(1,2)$.
   (a) At what rate is the y-coordinate of the point changing at that instant? Ans: $\frac{dy}{dt} = -\frac{60}{7}$
   (b) Is the particle rising or falling at that instant? Ans: Falling

9. A point P is moving along the curve whose equation is $y = \sqrt{x^3 + 17}$. When P is at $(2,5)$, y is increasing at the rate of 2 units/s. How fast is x changing? Ans: $\frac{dx}{dt} = \frac{1}{15}$ units/s

Tangent and Normal Lines

- The equation of the tangent to the curve $y = f(x)$ at (x_0, y_0)

$$y - y_0 = m\,(x - x_0) \qquad \text{where } m = \left(\frac{dy}{dx}\right)_{(x_0,y_0)}$$

- The equation of the normal at (x_0, y_0)

$$y - y_0 = -\frac{1}{m}(x - x_0), \text{ provided } m \neq 0.$$

Example-1: Find the equation of the tangent and normal to the curve $x^2 + 2xy + y^2 = 12$ at the point $(1,2)$.

Solution: Given, $x^2 + 2xy + y^2 = 12$

Differentiating with respect to x

$$2x + 2y + 2x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

or, $\frac{dy}{dx} = \frac{-2x-2y}{2x+2y}$

Now putting $x = 1, y = 2$

$$\left(\frac{dy}{dx}\right)_{(2,\,1)} = -\frac{6}{6}$$

or, $m = -1$

The equation of the tangent at (1,2):
$y - 2 = -1(x - 1)$
or, $y - 2 = -x + 1$
or, $x + y = 3$

The equation of the normal at (1, 2):
$y - 2 = -(-1)(x - 1)$
or, $y - 2 = x - 1$
or, $x - y = -1$

Example-2: Determine the equation of the tangent and normal to the curve given by $f(x) = 4x^3 + 12x^2 + 9x + 2$ at $x = -1$.

Solution: Given, $y = 4x^3 + 12x^2 + 9x + 2$

Putting $x = -1$,

$$f(x) = 4(-1)^3 + 12(-1)^2 + 9(-1) + 2$$
$$= -4 + 12 - 9 + 2 = 1$$

Therefore the tangent to the curve passes through the point $(-1,\ 1)$. Let us find out $f'(-1)$.

Now $f'(x) = 12x^2 + 24x + 9$

$f'(-1) = 12(-1)^2 + 24(-1) + 9 = 12 - 24 + 9 = -3.$

Therefore, the slope of the tangent to the curve at the point $(-1,\ 1)$ is $m = -3$.

The equation of the tangent at $(-1,1)$ is

$$y - 1 = -3(x - (-1))$$

$$\text{or, } y - 1 = -3(x + 1)$$

$$\text{or, } y - 1 = -3x - 3$$

$$\text{or, } y = -3x - 2$$

The equation of the normal at $(-1,1)$ is

$$y - 1 = -\frac{1}{-3}(x - (-1))$$

$$\text{or, } y - 1 = \frac{1}{3}(x + 1)$$

$$\text{or, } y = \frac{1}{3}x + \frac{4}{3}$$

Example-3: (P-655)#3 Find the equation of the tangent and normal to the curve at the given point.

$x = t^3 + 1, \quad y = t^4 + t; \quad t = 1.$

Solution Putting $t = 1$ we get $x_0 = 2$ and $y_0 = 2$.

Now, $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4t^3+1}{3t^2}$

or, $\left(\frac{dy}{dx}\right)_{t=1} = \frac{4(1)^3 + 1}{3(-1)^2} = \frac{4+1}{3} = \frac{5}{3}$

The equation of the tangent at (2,2):

$y - 2 = \frac{5}{3}(x - 2)$
or, $3y - 6 = 5x - 10$
or, $5x - 3y = 4$

The equation of the normal at (2, 2):

$y - 2 = -\frac{3}{5}(x - 2)$
or, $5y - 10 = -3x + 6$
or, $3x + 5y = 16$

Problem Set 3.2

1-11 Find the equations of the tangent line and normal line to the curve at the given point. (Ref: Calculus- James Stewart- 8th Edition. Page# 148, 149, 180, 215, 267, 655)

1. $y = 2x^3 - x^2 + 2,\ \ (1,3)$
2. $y = 2e^x + x,\ \ (0,2)$
3. $y = (2 + x)e^{-x}$
4. $y = x + \frac{2}{x},\ \ (2,3)$
5. $y = x^4 + 2e^x,\ \ (0,2)$
6. $y^2 = x^3,\ \ (1,1)$
7. $x^2 - xy - y^2 = 1,\ \ (2,1)$
8. $x^2 + 2xy + 4y^2 = 12,\ \ (2,1)$
9. $x^2 + 4xy + y^2 = 13,\ \ (2,1)$
10. $x = \sqrt{t}, y = t^2 - 2t,\ \ t = 4$
11. $x = t^3 + 1, y = t^4 + t,\ \ t = -1$

Taylor and Maclaurin Series

The function $f(x)$ has derivatives of all orders at $x = a$

- The Taylor series of $f(x)$ about $x = a$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^k(a)}{k!}(x-a)^k = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

- The Maclaurin series of $f(x)$ [i.e. The Taylor series of $f(x)$ at $x = 0$]

$$f(x) = \sum_{k=0}^{\infty} \frac{f^k(0)}{k!}x^k = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Example-1: Find the Taylor series for the function $f(x) = \frac{1}{x+1}$ at $x = 2$ using first three nonzero terms.

Solution: Given,

$$f(x) = \frac{1}{x+1}, \qquad f(2) = \frac{1}{3},$$

$$f'(x) = -\frac{1}{(x+1)^2}, \quad f'(2) = -\frac{1}{9},$$

$$f''(x) = \frac{2}{(x+1)^3} \qquad f''(2) = \frac{2}{27}$$

The Taylor series at $x = 2$ is,

$f(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \cdots$

or, $f(x) = \frac{1}{3} - \frac{1}{9}(x-2) + \frac{1}{27}(x-2)^2 + \cdots$

Example-2: Find the Maclaurin series for the function $f(x) = \cos x$ using first three nonzero terms.

Solution: Given, $f(x) = \cos x$, $f(0) = 1$,

$f'(x) = -\sin x$, $f'(0) = 0$,

$f''(x) = -\cos x$, $f''(0) = -1$

$f'''(x) = \sin x$, $f'''(0) = 0$

$f^{iv}(x) = \cos x$, $f^{iv}(0) = 1$

The Maclaurin series is,

$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{iv}(0)}{4!}x^4 + \cdots$

or, $f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$

Problem Set 3.3

1-5 Find the first three nonzero terms of the Taylor series for the following functions at the given value of x. (Ref: Calculus- James Stewart- 8th Edition. Page# 771)

1. $f(x) = xe^x,\ x = 0$
2. $f(x) = \sqrt[3]{x},\ x = 8$
3. $f(x) = \ln x,\ x = 1$
4. $f(x) = \sin x,\ x = \frac{\pi}{6}$
5. $f(x) = \cos^2 x,\ x = 0$

6-10 (Ref: Calculus- Howard Anton- 10th Edition. Page# 667)

6. $f(x) = \sin \pi x,\ x = \frac{1}{2}$
7. $f(x) = \cos x,\ x = \frac{\pi}{2}$
8. $f(x) = \ln x,\ x = e$
9. $f(x) = \frac{1}{x+2},\ x = 1$
10. $f(x) = \frac{1}{x},\ x = -1$

11-17 Find the first three nonzero terms of the Maclaurin series for the following functions. (Ref: Calculus- James Stewart- 8th Edition. Page# 771)

11. $f(x) = (1 - x)^{-2}$
12. $f(x) = \ln(1 + x)$
13. $f(x) = e^{-2x}$
14. $f(x) = 2^x$
15. $f(x) = x \cos x$
16. $f(x) = \sinh x$
17. $f(x) = \cosh x$

Chapter 4

Analysis of Function and Optimization Problems

Analysis of Function:

- **Stationary** point of a function $f(x)$ is a point where $f'(x) = 0$.
- **Inflection** point of a function $f(x)$ is a point where $f''(x) = 0$.
- Increasing or decreasing: Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b).
    - If $f'(x) > 0$ for every value of x in (a, b) , then $f(x)$ is increasing on $[a, b]$.
    - If $f'(x) < 0$ for every value of x in (a, b) , then $f(x)$ is decreasing on $[a, b]$.
- Concavity: Let f be twice differentiable on an open interval.
    - If $f''(x) > 0$ for every value of x in the open interval, then $f(x)$ is concave up on that interval.
    - If $f''(x) < 0$ for every value of x in the open interval, then $f(x)$ is concave down on that interval.
- Local Extrema (maximum or minimum):

  At a stationary point $x = c$, $f'(c) = 0$
    - If $f''(c) > 0$, then $f(x)$ has a local minimum at *c.*
    - If $f''(c) < 0$, then $f(x)$ has a local maximum at *c*.

Example 1: Consider the function $f(x) = x^3 + 3x^2 - 9x + 1$, then

(a) Find the stationary points.

(b) Find the interval(s) on which the function is increasing and decreasing.

(c) Find the points of inflection.

(d) Find the interval(s) on which the function is concave up and concave down.

(e) Discuss the nature of stationary points.

(f) Sketch the graph.

Solution: Given that $f(x) = x^3 + 3x^2 - 9x + 1$

(a) $f'(x) = 3x^2 + 6x - 9$

At the stationary points, $f'(x) = 3x^2 + 6x - 9 = 0 \Rightarrow x = -3,1$

Now, $f(-3) = 28$ and $f(1) = -4$. So, the stationary points are $(-3, 28)$ and $(1, -4)$.

(b) $f'(x) = 3x^2 + 6x - 9$

Intervals	Sign of $f'(x)$	Comment
$(-\infty, -3)$	positive	increasing
$(-3,1)$	negative	decreasing
$(1, +\infty)$	positive	increasing

So, $f(x)$ is increasing on $(-\infty, -3) \cup (1, +\infty)$ and decreasing on $(-3,1)$.

(c) $f''(x) = 6x + 6$

At the points of inflection, $f''(x) = 6x + 6 = 0, \Rightarrow x = -1$ Now, $f(-1) = 12$

So, the point of inflection is $(-1, 12)$.

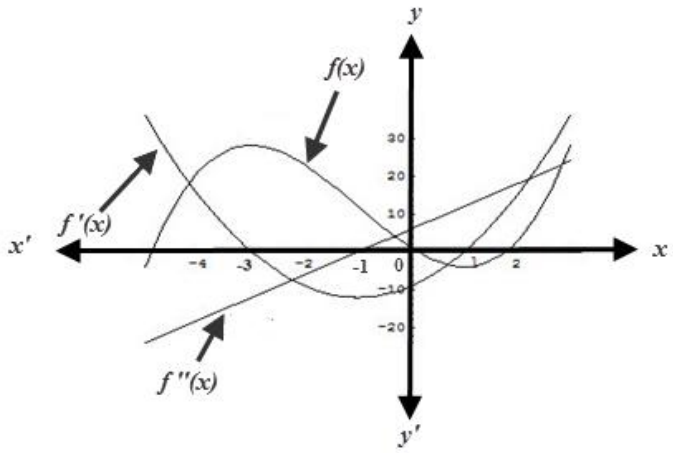
(d) $f''(x) = 6x + 6$

Intervals	Sign of $f''(x)$	Comment
$(-\infty, -1)$	negative	concave down
$(-1, +\infty)$	positive	concave up

So, $f(x)$ is concave up on $(-1, +\infty)$ and concave down on $(-\infty, -1)$.

(e) From the above figure, $f(x)$ has a local maximum at $(-3, 28)$ and local minima at $(1, -4)$.

(f)



Problem Set 4.1

(a) Find the stationary points.

(b) Find the interval(s) on which the function is increasing and decreasing.

(c) Find the points of inflection.

(d) Find the interval(s) on which the function is concave up and concave down.

(e) Discuss the nature of stationary points.

(f) Sketch the graph.

1. $f(x) = x^3 - 3x^2 - 9x + 4$
2. $f(x) = 2x^3 - 9x^2 + 12x - 3$
3. $f(x) = x^4 - 2x^2 + 3$
4. $f(x) = x^3 - 12x + 2$
5. $f(x) = 36x + 3x^2 - 2x^3$
6. $f(x) = \frac{1}{2}x^4 - 4x^2 + 3$
7. $f(x) = 200 + 8x^3 + x^4$ **[1-7]** [**Ref.** James Stewart 8th edition, page-301,302]
8. $f(x) = x^3 - 9x + 1$
9. $f(x) = 2x^3 - 6x + 2$
10. $f(x) = 3 - 4x - 3x^2$ **[8-10]** [**Ref.** Howard Anton-10th edition, page-273]

Optimization Problems:

- The problem of finding the best solution from all feasible solutions
- To minimize or maximize a function subject to given constraint(s).

Example 1: A garden is to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if and only if 100 running feet of chicken wire is available for the fence?

Solution: Let,

$x =$ length of the rectangle (ft)

$y =$ width of the rectangle (ft)

$A =$ area of the rectangle (ft per square)

Then, we know $A = xy$

Since the perimeter of the rectangle is 100 ft, the variables x and y are related by the equation,

$2x + 2y = 100$ or $y = 50 - x$.

Now the area, $A = x(50 - x) = 50x - x^2$.

Because x represents a length, it can not be negative, and because the two sides of length x cannot have a combined length exceeding the total perimeter of 100 ft, the variable x must satisfy, $0 \leq x \leq 50$.

Thus, we have reduced the problem to that of finding the value of x in $[0,50]$, for which A is maximum. Since A is a polynomial in x, it is continuous on $[0,50]$ and so the maximum must occur at an endpoint of the interval or at a critical point.

$$\frac{dA}{dx} = 50 - 2x \; or \; \frac{dA}{dx} = 50 - 2x = 0$$

$$x = 25$$

Putting the values of x in area we get the maximum value at $x = 25$ which is $625\ ft^2$

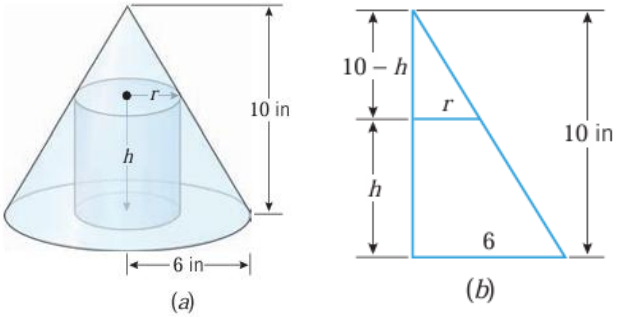
Example 2: Find the radius and height of the right circular cylinder garden of largest volume that can be inscribed in a right circular cone with radius 6 inches and height 10 inches.

Solution: Let,

$r =$ radius of the cylinder (inches)

$h =$ height of the cylinder (inches)

$V =$ volume of the cylinder (cubic inches)



Then, we know $V = \pi r^2 h$

Using similar triangle, we obtain,

$\frac{r}{6} = \frac{10-h}{10} \Rightarrow h = 10 - \frac{5}{3}r$

So, $V = \pi r^2 \left(10 - \frac{5}{3}r\right) = 10\pi r^2 - \frac{5}{3}\pi r^3$

Here, $0 \leq r \leq 6$ thus we have reduced the problem to that of finding the value of r in $[0,6]$ such that V is a maximum.

$$\frac{dV}{dr} = 20\pi r - 5\pi r^2 = 5\pi r(4 - r)$$

Now,

$$\frac{dV}{dr} = 0$$

$$\Rightarrow 5\pi r(4 - r) = 0$$

$$\therefore r = 0, 4$$

Putting the values of r in volume we get the maximum volume $V = \frac{160}{3}\pi \approx 168$ in^3 occurs when the inscribed cylinder had radius 4 in. When $r = 4$ in, it follows that $h = \frac{10}{3}$ in.

Problem Set 4.2

1. A farmer has $2400\ ft$ of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

2. A cylindrical can is to be made to hold $1\ L$ of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

3. Find the area of the largest rectangle that can be inscribed in a semicircle of radius r.

4. If $1200\ cm^2$ of material is available to make a box with a square base and an open top, find the largest possible volume of the box.

5. A box with a square base and open top must have a volume of $32{,}000\ cm^3$. Find the dimensions of the box that minimizes the amount of material used.

6. A rectangular storage container with an open top is to have a volume of $10\ m^3$. The length of its base is twice the width. Material for the base costs $\$10$ per square meter. Material for the sides costs $\$6$ per square meter. Find the cost of materials for the cheapest such container.

[1-6] [**Ref.** James Stewart 8th edition, page-331, 337]